

OBJECTIVE:

<OPTIONAL: 1 sentence about the specific position/company (name it) and your interest/knowledge that makes you most relevant>

EDUCATION:

Doctor of Philosophy in Computer Science
Clemson University

May 2025
Clemson, SC

Bachelor of Science in Computer Science with Minor in Business Administration
Clemson University

May 2019
Clemson, SC

RELEVANT EXPERIENCE:

EXPERIENCE:

Volunteer Research Assistant
Clemson University

Fall 2025—Present
Clemson, SC

- Cross-lab/org collaborations
- Project Lead

Graduate Research Assistant
Clemson University

Summer 2019—Spring 2025
Clemson, SC

- Analyzed thermal behaviors for High Performance Computing clusters in Single Phase Liquid Immersion Cooling environments with novel data harness
- Optimized the Word2Vec Neural Network for NVIDIA Pascal and Volta architectures, resulting in 472% higher throughput while improving between-architecture scaling by 197%
- Group meetings
- Cross-lab/org collaborations
- Lab mentorship
- Published

Vice President

Fall 2024—Spring 2025
Clemson, SC

School of Computing Graduate Student Association

- Organized 3 professional workshops, 3 social events and a fundraiser, increasing student engagement
- Managed organization finances, plans, marketing and personnel to revitalize organization post-COVID

OMNI Internship Program, Graduate Student Researcher

Summer 2024

Oak Ridge National Laboratory

Oak Ridge, TN

- Identified key insights on LLM in-context learning and inference behaviors that limit current capabilities to generalize across new data without additional training or fine-tuning
- Mentoring other students in program
- Published work

Lead Student Volunteer, Guided Interest Group Leader, HPC Immersion Mentor

Fall 2023, 2024

SuperComputing '23, '24

Denver, CO; Atlanta, GA

- Established Reproducibility efforts and ACM Badge Verification processes for the SuperComputing conference
- Organized over 30 conference Panels and Birds of a Feather with conference committee members
- Managed 100+ student volunteers to facilitate conference events and coordinated efforts serving up to 15,000 guests
- Developed conference attendance plans and paper discussion sessions for groups of three to seven students, focusing on encouraging and empowering students to make the most of their conference attendance while fostering a learning environment
- Formed individual mentoring relationships in one-on-one settings with newcomers to the High Performance Computing field, ensuring they were welcomed and involved in their conference experience

GIVENS Scholar, Graduate Student Researcher

Summer 2020, 2022, 2023

Argonne National Laboratory

Lemont, IL

- Created a novel methodology for automatic performance tuning via transfer learning with probabilistic performance modeling that achieves 33.39x additional speedup over prior state-of-the-art techniques
- Mentoring other students
- Published work

Graduate Teaching Assistant

Fall 2019 — Spring 2023

Clemson University, School of Computing

Clemson, SC

- Prepared assignments, projects and homework for classes with 30-100+ students (up to 8 hours per week)
- Aided students in office hours with questions about course materials and assignments (3-5 hours per week)
- Developed a new course for Clemson University with Assistant Professor Dr. Mert Pesé
- Delivered consistent and actionable feedback to students within 7 days of submission on labs, projects, quizzes and exams

Student Volunteer, Virtual Student Volunteer

Fall 2020 (Virtual), Fall 2021, Fall 2022

SuperComputing'20, SuperComputing'21, SuperComputing'22

Online; St. Louis, MO; Dallas, TX

- Moderated online discussions and forward questions for panels and technical presenters with over 1000 live members in the audience

- Facilitated chain-of-command communications between conference organizers, volunteers, building staff and event security, improving response times to emergency situations
- Undergraduate Research Assistant Summer 2018—Spring 2019
Clemson University Clemson, SC
- Optimized a CPU kernel to accelerate Word2Vec kernel performance, achieving 1.3x speedup over prior state-of-the-art throughput
 - Performed GPU kernel performance analysis and development, identifying key bottlenecks and possible algorithmic remedies to address performance scaling issues
 - Assimilated in-depth background research and algorithmic analysis of full-scale machine learning application
 - Created and delivered presentations to demonstrate learning progress and setbacks for weekly group presentations
- Laboratory Teaching Assistant Spring 2017 — Spring 2019
Clemson University, School of Computing Clemson, SC
- Prepared and introduced labs and projects for up to 30 students (up to 8 hours per week)
 - Delivered consistent and actionable feedback to students with 7-day turnarounds on submitted labs and projects
 - Aided students with questions and problems during lab time and office hours (3-5 hours per week)
- Summer Staff Coordinator Summer 2017
Isaiah 55 Deaf Ministries La Feria, TX
- Coordinated and supervised volunteer teams of 5-15 on small-scale construction and renovation projects, completing 30 projects over a 10-week summer term
 - Organized daily community outreach events including VBS and children's arts & crafts and recreation, serving 10-100 children per day
 - Managed logistics for tool distribution and construction materials to reduce downtime setting up and closing down workdays

SKILLS:

ENGAGEMENT (Communication, Collaboration, Leadership)

- Scientific writing
- Conference presentations (public speaking)
- Mentoring graduate and undergraduate students

INNOVATION (Adaptability, Analysis, Technology)

- Scientific research and analysis synthesis
- Performance modeling
- Performance optimization
- High Performance Computing
- C, CUDA, Python

Professionalism (Self-Awareness, Integrity, Ethics, Brand)

- Time management
- Self-motivated
- Learn from feedback

AWARDS:

Outstanding PhD Student in Computer Science

Spring 2025

ICS'21 Best Paper

Summer 2021

R.C. Edwards Graduate Fellow

Fall 2019—Spring 2022

Clemson School of Computing Outstanding Undergraduate Researcher

Spring 2019

Dupont Best Undergraduate Project

Spring 2019

INVITED TALKS AND FEATURED COVERAGE:

Oak Ridge National Lab Artificial Intelligence Seminar Series

Spring 2024

Invited Presenter

Oak Ridge, TN

Argonne National Lab Computer Science Seminar Series

Summer 2023

Invited Presenter

Lemont, IL

I Am HPC

Spring 2023

<https://sc23.supercomputing.org/2023/05/hpc-on-the-rise-thomas-randall/>

Online

Ask A Grad

Fall 2022

SoC-GSA & UPE

Clemson, SC